2023년 겨울 학기

Digital Logic Design – Digital Signals

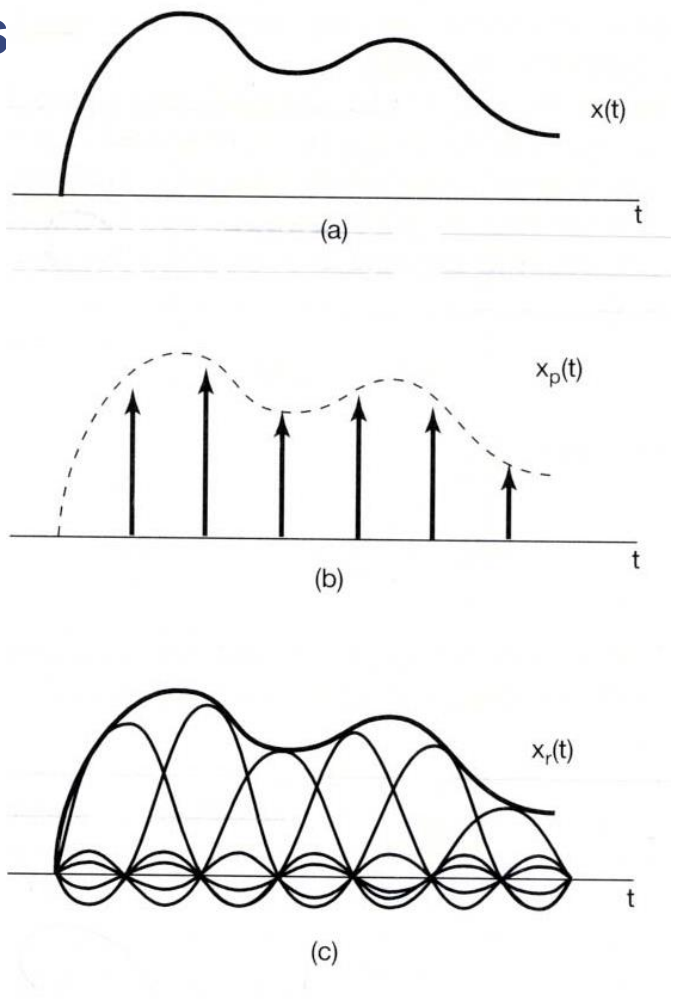
이 상 화

Signals

- **Signal is a kind of function**
  - Time, spatial variables
- **Physical quantity and entity**
  - Voltage, current, sensor outputs, group of data (vector)
- **Signal Types**
  - Analog signal
    - Continuous values
    - Audios, displayed images/videos
  - Discrete signal
    - Obtained by sampling analog signals
  - Digital signal
    - Quantization of signal values and binarization

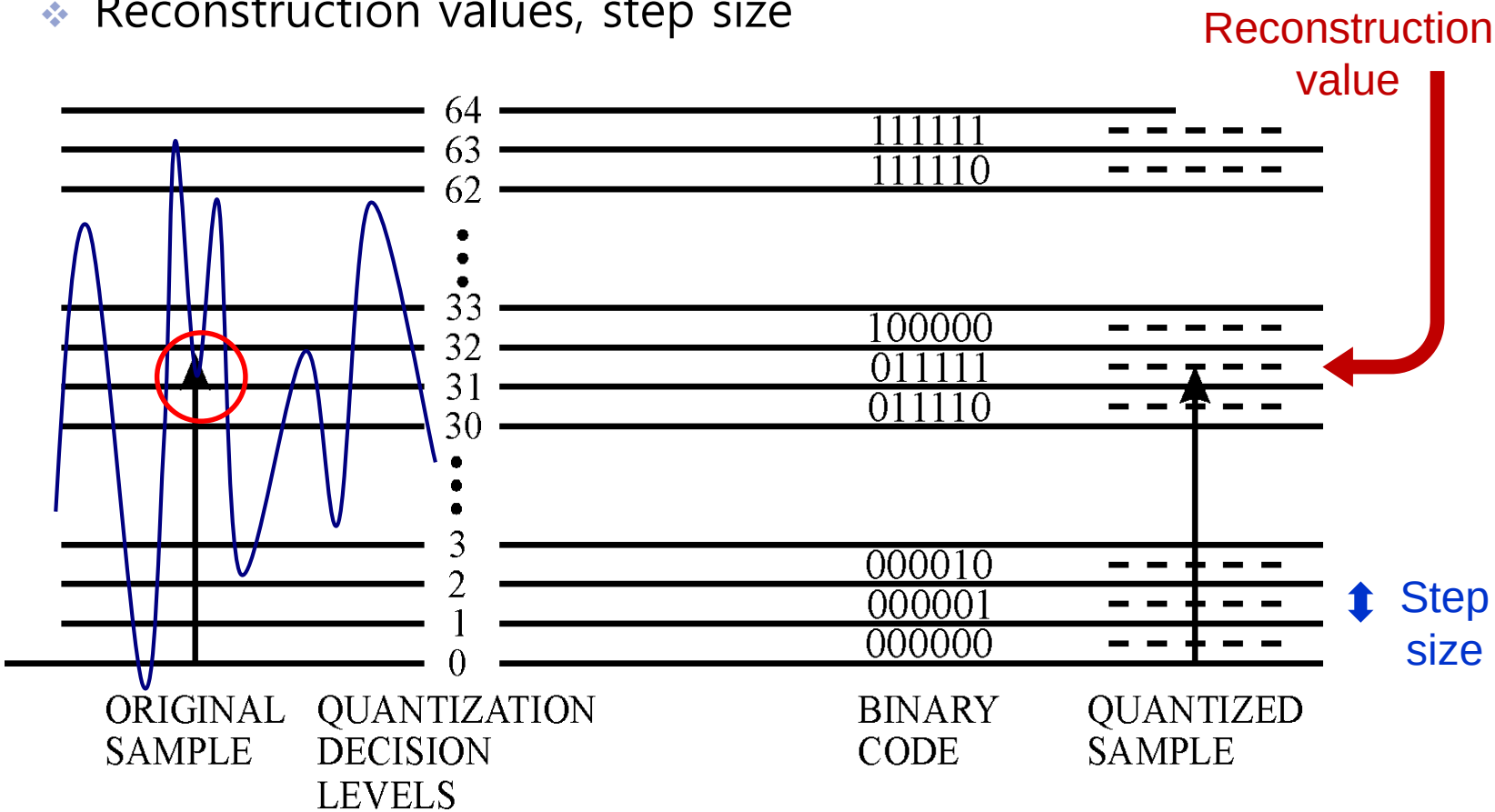
Sampling (표본화)

- **Obtaining discrete (이산) signals from analog (연속) signals**
  - (a) Analog (audio) signal
  - (b) sampling with time period T
  - (c) Reconstruction of original analog (continuous) signal using Low pass filtering
- **Example**
  - Audio signals
    - 20~20kHz
    - Sampling with 44.1kHz
  - Nyquist sampling
    - $f_s > 2 f_{max}$



Quantization (양자화) (1)

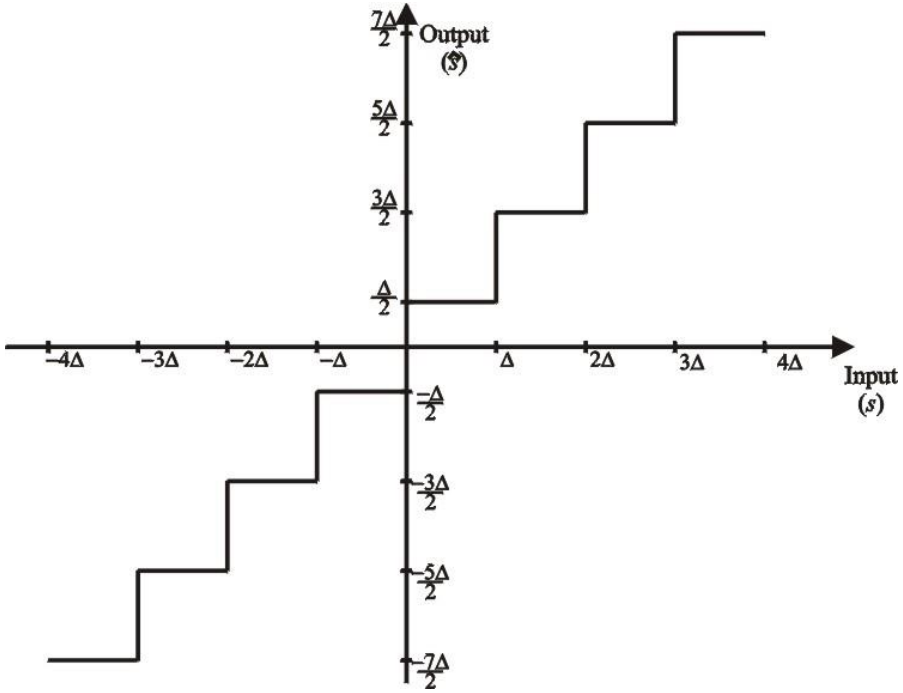
- 무한한 레벨로 표현되는 연속적인 값을 유한한 개수의 **discrete value**를 갖는 **level**로 표현하는 과정
  - Data clustering
  - Reconstruction values, step size



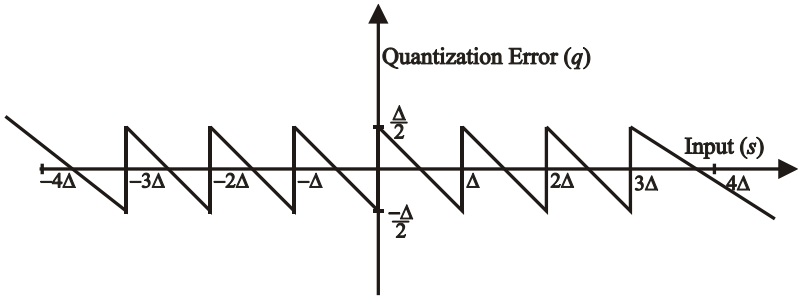
Quantization (2)

- **Quantization error (e)**

$$e = Q(x) - x$$



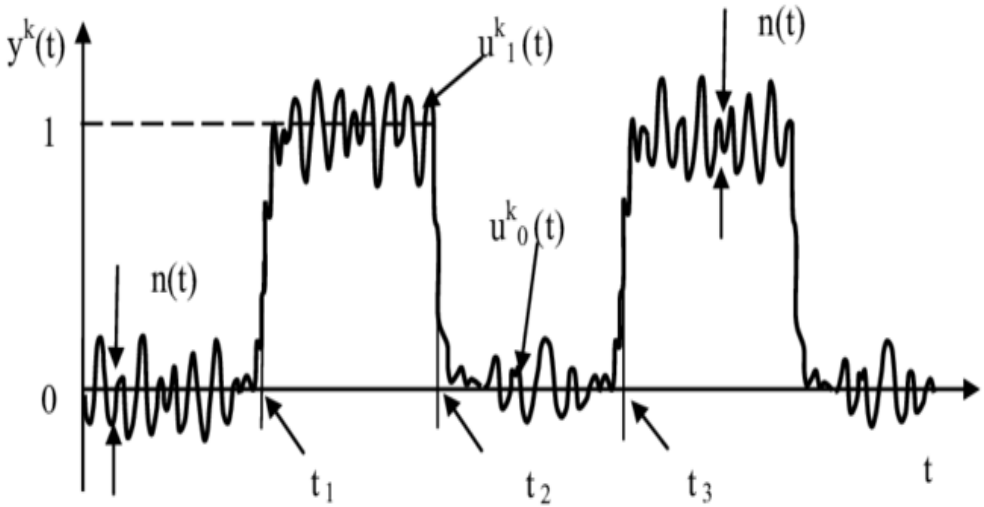
Uniform quantizer



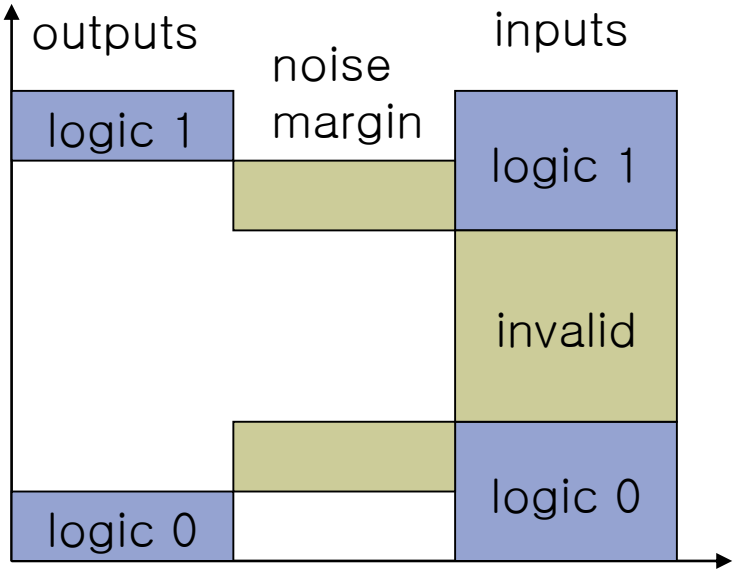
Uniform distribution
of quantizer error

Analog vs Digital

- **Decision of digital 0 and 1**
  - Considering noise in real physical signals



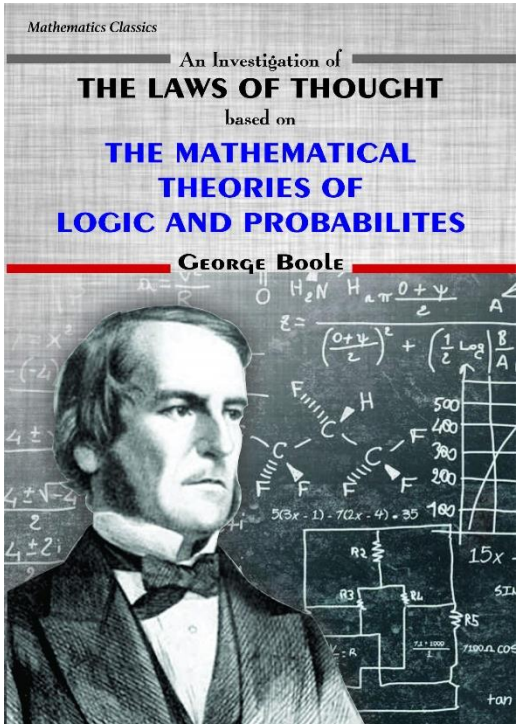
Real digital signal

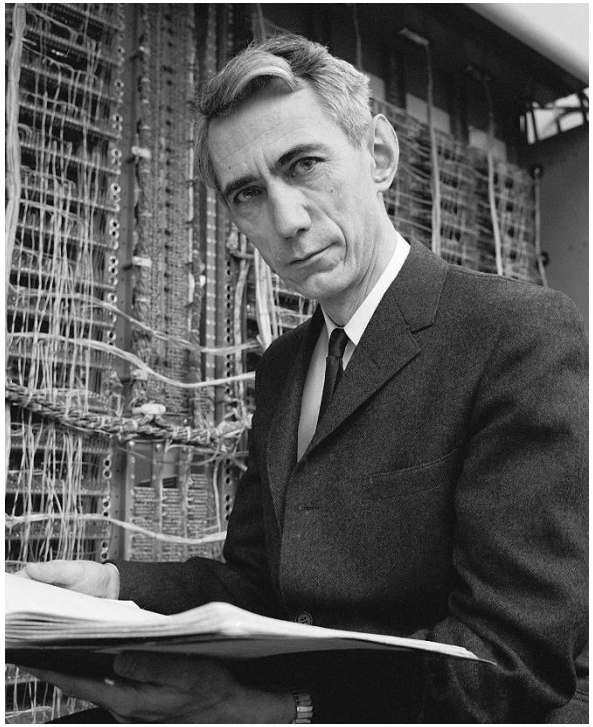


Binary decision from real signal

Quick History (1)

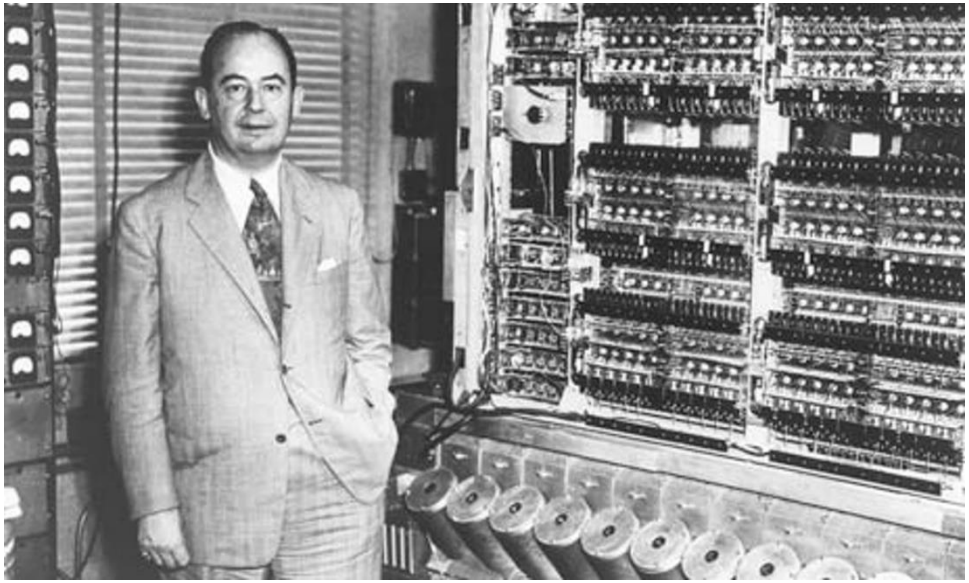
- **1850: George Boole invents Boolean algebra**
  - maps logical propositions to symbols
  - permits manipulation of logic statements using mathematics
- **1938: Claude Shannon links Boolean algebra to switches in his Masters' thesis**





Quick History (2)

- **1945-1952: John von Neumann develops the first stored program computer**
  - its switching elements are vacuum tubes (a big advance from relays)
  - 1946: ENIAC, The first completely electronic computer
    - 18,000 vacuum tubes



Von Neumann

앨런 튜링이 암호 해독을 위해 1943년에 만든 컴퓨터 '콜로서스'

Quick History (3)

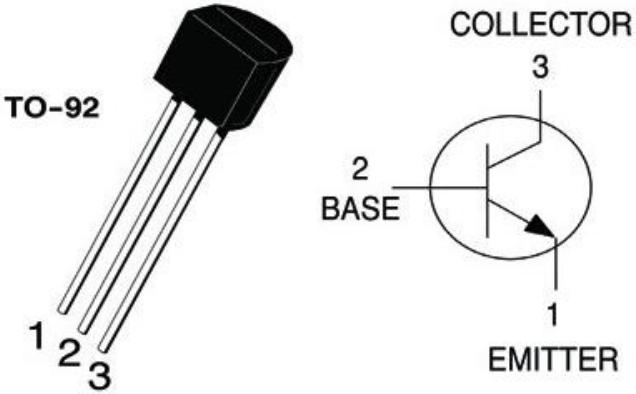
- **1947: Shockley, Brittain, and Bardeen invented the transistor**
  - replaces vacuum tubes
  - enable integration of multiple devices into one package
  - gateway to modern electronics

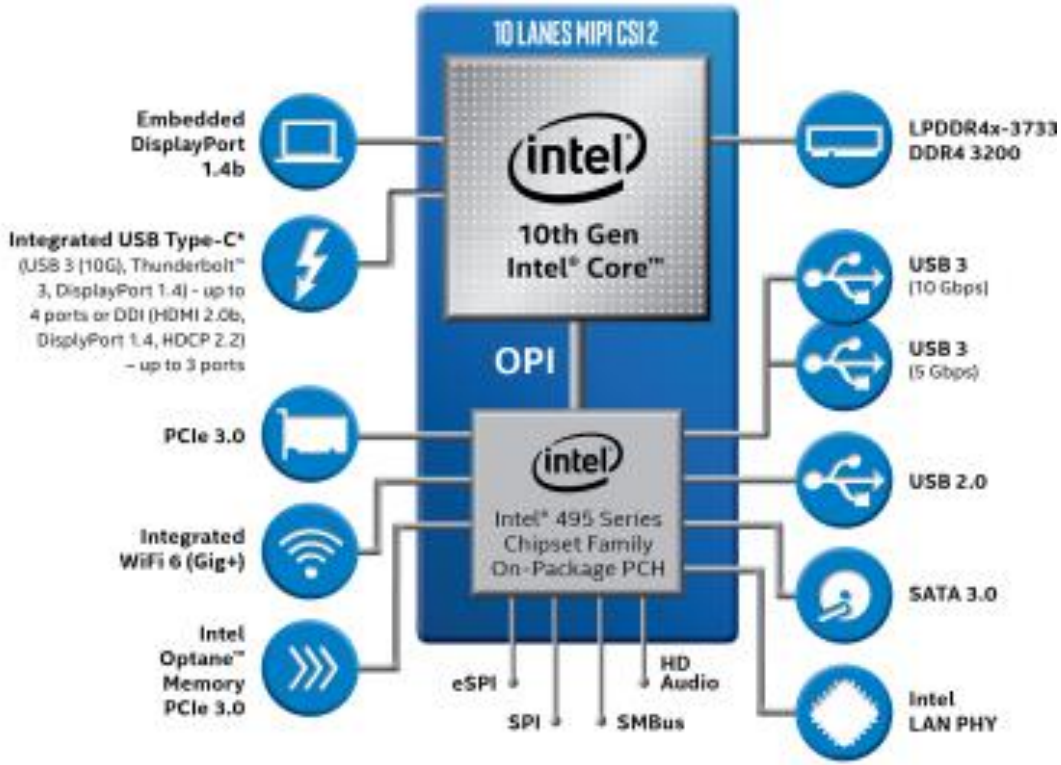
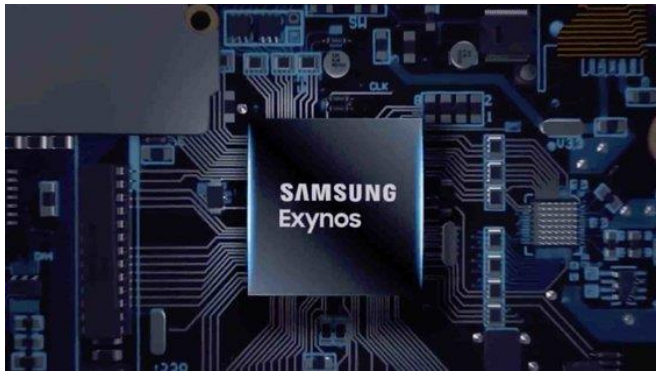
The Nobel Prize in Physics 1956

William Bradford Shockley
Prize share: 1/3

John Bardeen
Prize share: 1/3

Walter Houser Brattain
Prize share: 1/3



Quick History (4)

- **2020'**
  - CPU, GPU, multimedia processors chip set
  - Integration of billions of transistors